



ATOMARINE

Nuclear Propulsion as a Service

Decarbonizing global shipping with detachable offshore nuclear power.

Dimitris Koutentakis | Emile Germonpré

The Dirtiest Fuel on Earth

Ships burn heavy fuel oil: a thick, black, sludge-like residual so toxic it must be heated just to flow.



3%

of global CO₂ emissions

Shipping would be the 6th largest emitter if it were a country

100K+

premature deaths / year

From SO_x and NO_x pollution caused by burning HFO

\$127B

annual fuel spend

With extreme price volatility—spot prices can double in months

Ships Are Halving Their Speeds

To meet emissions targets, the industry is slow-steaming: Cutting speeds by up to 50%. This slows global deliveries, erodes freight revenue, and still doesn't solve the carbon problem.

50%

speed reduction mandated

CII ratings force vessels to slow-steam or face penalties. Global supply chains are paying the price.

10×

fuel cost increase by 2045

EU ETS + FuelEU Maritime carbon penalties compound year over year. Non-compliance is existential.

Nuclear Checks Every Box. Nothing Else Does.

✗ **Ammonia / H₂**

No scalable supply chain. Unproven at scale. 3–5× fuel cost.

✗ **Wind / Sails**

Supplemental only. Cannot propel deep-sea vessels independently.

✗ **Biofuels**

Land-use competition. Limited supply. Not fully carbon-neutral.

✗ **Carbon Capture**

Unproven at maritime scale. Adds weight, cost, and complexity.

N U C L E A R

- ✓ Scalable supply chain
- ✓ Proven: 70+ years at sea
- ✓ Cost-effective at scale
- ✓ 100% zero-emission
- ✓ Maintains full speed

But conventional nuclear shipping has been blocked for 60 years.

Why Nuclear Shipping Has Failed for 60 Years

Every attempt puts a reactor on a cargo ship. Every attempt hits the same four walls.

1

Port Access

Nuclear ships denied entry to commercial ports worldwide. Emergency zones around population centers make it impossible.

2

Liability

Billions in nuclear liability per vessel. No shipowner will take this on their balance sheet.

3

Regulation

Every port state demands separate licensing. One voyage crosses 5+ regulatory regimes.

4

Public Opposition

Reactors near populated ports face intense resistance. The political cost is prohibitive.

All four barriers share one root cause: the reactor enters port.

Detach the Reactor.

Stay Offshore. Power the Fleet.

Nuclear LeadShips

power convoys of commercial ships via dynamic offshore cables

Never enter port

operate exclusively in international waters

Zero modifications

for ships with electric drive. \$2–3M retrofit for direct-drive.

Cargo-agnostic

one standardized design serves all vessel types



99%

of voyage under nuclear power

Up to 6

ships per convoy

<5 yr

LeadShip payback

60 yr

LeadShip design life

Propulsion-as-a-Service, Per Voyage

01

DEPART

Ships leave port
under own power

02

RENDEZVOUS

Meet LeadShip
beyond EEZ

03

CONNECT

Dynamic cables
link to convoy

04

TRANSIT

Nuclear convoy
crosses at 15+ kts

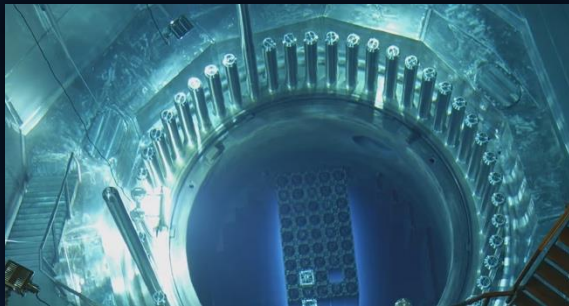
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DISCONNECT

Ships detach,
enter port

Ships currently burning dirty HFO switch to clean nuclear power for 99% of the voyage and achieve high speed without carbon penalties.

No New Physics. Three Proven Pillars.



Marine iPWR Reactors

160+ naval vessels • 2,000+ reactor-years

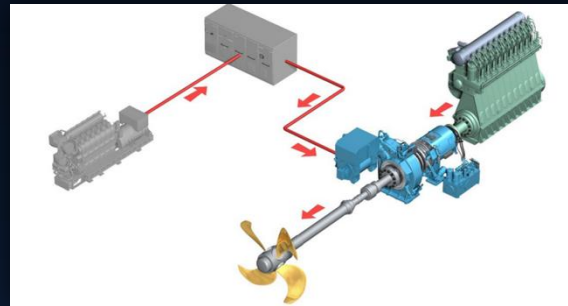
Integral PWR with 70+ years of maritime heritage. Modern SMR designs already in licensing.



Dynamic Offshore Cables

Proven in floating wind & oil/gas

High-voltage dynamic power cables rated for open-ocean conditions. Multiple commercial manufacturers.



Electric-Drive Propulsion

Thousands of ships already equipped

PTO/PTI systems standard in LNG carriers, cruise ships, icebreakers. Retrofit cost: \$2–3M.

One Insight. Every Barrier Resolved.

X Port Access Denied



✓ LeadShips never enter port - international waters only

X Liability & Insurance



✓ Zero nuclear liability on shipowner balance sheets

X Multi-Jurisdiction Regs



✓ Single Flag State licensing: one design, one jurisdiction

X Public Opposition



✓ No reactors near population centers

X High CAPEX per Ship



✓ Shared across convoys - infrastructure economics

X Crew per Ship



✓ Centralized on LeadShip: no nuclear training for clients

Every competitor puts reactors on cargo ships. We removed the reactor from the ship entirely.

Infrastructure Economics, Not Shipping Economics

REVENUE MODEL

Pricing

Per-voyage fee indexed to energy + avoided carbon penalties

Contracts

Long-term take-or-pay secured by EU carbon tax savings

Revenue

\$150–250M annually per LeadShip at >90% utilization

Margins

Staffing doesn't scale with power — convoys of 4+ unlock strong margins

WHY SHIP OPERATORS BUY

- ✓ Total cost < fuel + carbon tax
- ✓ Zero nuclear liability
- ✓ No specialized crew needed
- ✓ Restore full sailing speed
- ✓ Compliance credits banked & sold
- ✓ Minimal retrofit (~\$2–3M PTI)

A \$127B+ Market Under Existential Pressure

TAM

\$127B+

Annual global shipping fuel spend
+ \$1–100B in FuelEU carbon costs

SAM

\$95B

Bulk carriers + tankers on routes >1,000 NM
(75% of fuel consumption)

SOM \$1.5B+

Container ships on EU-regulated routes

BEYOND MARITIME

The offshore nuclear infrastructure model extends to:

- ✓ **Floating Datacenters**
Centrally build self-powered datacenters and ship to cool locations
- ✓ **Remote energy supply**
Plug in to any coastal grid
- ✓ **Industrial decarbonization**
Provide carbon-free energy to desalination plants
- ✓ **Defense & logistics**
Charge electric autonomous vessels, power deployments, provide emergency power

Convoys Unlock Economies of Scale

\$1.0–1.6B

CAPEX

Hull + reactor + NRE

60 yr

DESIGN LIFE

2× merchant ship lifespan

\$150–250M

ANNUAL REVENUE

Per LeadShip at >90% util.

PAYBACK BY SCENARIO

2045 Carbon + 4 ships

5 yr

2040 Carbon + 3 ships

8 yr

2040 Carbon + 2 ships

12 yr

Conservative (1 ship)

20 yr

22% learning rate. At fleet scale (50 LeadShips), per-unit CAPEX drops by >60%.

The Only Architecture That Bypasses the Barriers

COMPANY	APPROACH	PORT ACCESS	TIMELINE	SCALABLE
Allseas	Onboard SMR	Required	Long	No
Core Power	Floating NPPs	Required	Long	Partial
Maritime Fusion	Compact fusion	Required	Very Long	Unproven
SMR Vendors	Near-shore float	Required	Medium	Partial
Atomarine	Offshore NPaaS	Eliminated	Medium	Yes

Atomarine is faster, cheaper, and eliminates all risk for the shipowner.

Five Layers of Compounding Moat

- 1 Regulatory Lock-in** First Flag State nuclear code creates path dependency
- 2 Network Effect** More LeadShips => shorter wait times => better service
- 3 Technology Lock-in** Ships designed around our connectors. High switching costs.
- 4 Intellectual Property** Provisional patents filed. Multi-jurisdiction filings planned.
- 5 Talent Moat** First to assemble nuclear-maritime cross-disciplinary team.

TRACTION

Industry Commitments and Validation

SIGNED LETTERS OF SUPPORT

X-Energy

Leading SMR developer - reactor supply partnership

Apollo Atomics

Advanced nuclear technology - strategic collaboration

INDUSTRY COLLABORATIONS

Costamare

Information exchange & operations validation

MPC Container Ships

Operational validation partnership

Dorian LPG

Fleet compatibility assessment

ACADEMIC & INSTITUTIONAL

MIT / The Engine — Blueprint program

ABS — Approval in Principle pathway

Exmar — Shipping & LNG industry partner

Fulbright Program — Research fellowship

DNV — Classification (in discussions)

Multiple SMR Vendors — Reactor supply

TEAM

Expertise and Market Access



Dimitris Koutentakis

Co-Founder & CEO

MIT MBA & Mechanical Engineering MS
MIT EECS Undergrad & MEng
Amazon operations optimization



Emile Germonpré

Co-Founder & CTO

MIT Nuclear Science & Engineering PhD
UGent Chemical Eng. (Valedictorian)
Computational physics & risk analysis
Fulbright Scholar

KEY PARTNERS



Staged De-Risking to First Commercial LeadShip



Raising \$2-5M Pre-Seed Round

USE OF FUNDS



Engineering Team



FEED & AiP



At-Sea Demos



Regulatory & IP



Biz Dev

KEY MILESTONES (12 MO)

- ✓ Grow team
- ✓ Regulatory scoping with class societies
- ✓ Initiate AiP with ABS
- ✓ Ship-to-ship HV power transfer demo
- ✓ 3+ non-binding LOIs
- ✓ Select reactor vendor
- ✓ File non-provisional patents

The Post-Combustion Era Starts Now

1

Regulatory Certainty

EU ETS and FuelEU make carbon-free shipping inevitable. HFO's volatile pricing and compounding penalties create an untenable cost structure.

2

Technology Readiness

SMR licensing accelerating globally. Dynamic cables proven. Electric-drive ships proliferating. Every component exists today.

3

No Alternative Path

Every green fuel has failed to scale. Speed mandates are slowing global trade. Nuclear is the only energy source that checks every box.



ATOMARINE

The next century of global shipping is nuclear.

Atomarine is building the offshore nuclear infrastructure that makes it possible.

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